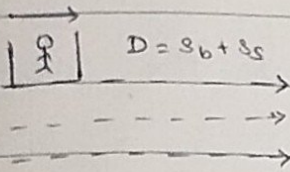


BOATS AND STREAMS

1.



(D) Downstream $\rightarrow$ if
person/boat goes along
(in the direction)
with stream.

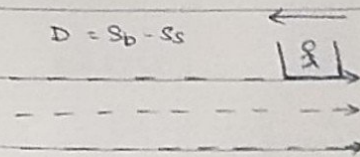
$$D = S_b + S_s$$

$$U = S_b - S_s$$

(+)

$$D + U = 2S_b$$

$$S_b = \frac{D + U}{2}$$



upstream (U) $\rightarrow$ if
person/boat is going
against the stream

$$D = S_b + S_s$$

$$U = S_b - S_s$$

(-)

$$D - U = 2S_s$$

$$S_s = \frac{D - U}{2}$$

$S_b \rightarrow$ Speed of boat in
still water

$S_s \rightarrow$ Speed of stream

$U \rightarrow$ upstream

$D \rightarrow$ downstream

Speeds kmph is the unit.

1. If a boat goes 40 km upstream in 8 hrs & 36 hrs downstream in 6 hrs. Find the speed of boat in still water.

Ans: 5.5 kmph

40 km 8 hrs
└──────────┘
upstream

36 km 6 hrs
└──────────┘
downstream

$$U = \frac{40}{8} = 5 \text{ kmph}$$

$$D = \frac{36}{6} = 6 \text{ kmph}$$

$$S_b = \frac{D+U}{2} = \frac{5+6}{2}$$

$$S_b = 5.5 \text{ kmph}$$

2. If a man's speed with ^{stream} current is 15 kmph, and speed of the current is 2.5 kmph. Find the man's speed against the current.

$$\text{Ans: } 10 \text{ kmph}$$

$$D = 15 \text{ kmph}$$

$$S_s = D - U$$

$$S_s = 2.5 \text{ kmph}$$

$$2.5 = \frac{15 - U}{2}$$

$$U = ?$$

$$U = 10 \text{ kmph}$$

3. The speed of the boat in still water is 27 kmph. It covers an upstream distance of 85 km in 4 hrs 15 mins. Find the speed of the stream.

$$\text{Ans: } 7 \text{ kmph}$$

$$S_b = 27 \text{ kmph}$$

$$85 \text{ km, } 4 \text{ hr } 15 \text{ min}$$

$$\text{Speed (U)} = \frac{85}{(4 \frac{15}{4})} = 85 \times \frac{4}{17}$$

upstream

$$4 + \frac{15}{60} = 4 + \frac{1}{4} = 4 \frac{1}{4} \text{ hrs}$$

$$= 20 \text{ kmph}$$

$$U = S_b - S_s \Rightarrow S_s = S_b - U \Rightarrow 27 - 20$$

$$U = 7 \text{ kmph}$$

4. Time taken by a boat to travel upstream is twice the time taken to travel downstream. If the speed of the boat in still water is 9 kmph. find the speed of the stream.

Ans: 3 kmph

$$t_u = 2 t_d$$

$$S \propto \frac{1}{t}$$

$$D = x \text{ kmph}$$

$$S_b = 9, S_s = ?$$

$$U = \frac{x}{2} \text{ kmph}$$

$$S_b = \frac{D+U}{2}$$

Time is doubled,

$$D = x \text{ kmph}$$

$$D = 12 \text{ kmph}$$

$$U = \frac{x}{2} \text{ kmph}$$

$$U = 6 \text{ kmph}$$

$$\frac{D+U}{2} = 9 \Rightarrow D+U = 18$$

$$x + \frac{x}{2} = 18$$

$$\frac{3x}{2} = 18$$

$$x = 12$$

$$S_s = \frac{D-U}{2} = \frac{12-6}{2} = 3 \text{ kmph}$$

5. If speed of boat in still water is 6 kmph and speed of stream is 1.5 kmph. A man rows to a place at a distance of 22.5 km and comes back to the starting point. Find the total time taken by him.

Ans: 8 hrs

$$t = t_u + t_D$$

$$t = \frac{22.5}{4.5} + \frac{22.5}{7.5}$$

$$= \frac{22.5}{U} + \frac{22.5}{D}$$

$$t = 5 + 3$$

$$t = 8 \text{ hrs}$$

$$U = s_b - s_s = 4.5 \text{ kmph}$$

$$D = s_b + s_s = 7.5 \text{ kmph}$$

- 6) The speed of motor boat to that of speed of current is in the ratio 36:5. The boat goes along the current in 5 hr 10 min. In what time it will come back.

$$\text{Ans: } 6 \text{ hrs } 50 \text{ min}$$

$$s_b : s_s = 36 : 5$$

$$s_b = 36x$$

$$U = s_b - s_s = 31x$$

$$s_s = 5x$$

$$D = s_b + s_s = 41x$$

$$U : D = 31 : 41 \quad [\text{speed ratio}]$$

$$t_u : t_D = 41 : 31$$

$$\begin{array}{ccc} 41 & & 31 \\ \downarrow & & \downarrow \\ 6 \text{ hrs } 50 \text{ min} & & 5 \text{ hr } 10 \text{ min} \end{array}$$

- 7) A man goes downstream with a boat and returns upstream to the starting point in 5 hrs. If the speed of boat in still water and speed of the stream is 10 kmph & 4 kmph respectively.

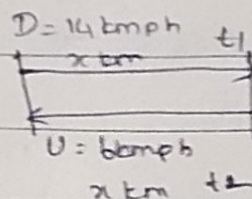
Find the distance between starting point and destination.

Ans: 21 km

$$S_b = 10 \text{ kmph} \quad S_s = 4 \text{ kmph}$$

$$D = S_b + S_s = 14 \text{ kmph}$$

$$U = S_b - S_s = 6 \text{ kmph}$$



$$t = t_1 + t_2$$

$$5 = \frac{x}{14} + \frac{x}{6}$$

$$\frac{x}{21} = 1$$

$$x = 21 \text{ km}$$

8) A person takes row to a distance of $3\frac{3}{4}$ km upstream in $1\frac{1}{2}$ hrs and a distance of 13 km downstream in 2 hrs. How much time will the person takes to row a percent of 90 km in still water.

Ans: 20 hrs

$$U = \frac{15/4}{3/2} = \frac{15}{4} \times \frac{2}{3} = 5/2 \text{ kmph}$$

$$D = 13/2 \text{ kmph}$$

$$S_b = \frac{D+U}{2} = \frac{13/2 + 5/2}{2} = \frac{9}{2} = 4.5 \text{ kmph}$$

$$4.5 \text{ kmph} \rightarrow 1 \text{ hr}$$

$$\downarrow \times 20$$

$$90 \text{ km}$$

$$\downarrow \times 20$$

$$20 \text{ hrs}$$

9) A motor boat whose speed is 15 kmph in still water goes 30 km downstream and comes back in 4 hrs 30 min. Find the speed of the stream.

Ans: 5 kmph

$$S_b = 15 \text{ kmph}$$

$$d = 30 \text{ km}$$

$$t = 4 \text{ hrs } 30 \text{ min}$$

$$S_s = x \text{ kmph}$$

$$U = S_b - S_s = 15 - x$$

$$D = S_b + S_s = 15 + x$$

$$T = t_U + t_D$$

$$\text{(4 hrs 30 min)} \frac{9}{2} = \frac{30}{15-x} + \frac{30}{15+x}$$

$$= 30 \left(\frac{1}{15-x} + \frac{1}{15+x} \right)$$

$$= 30 \left(\frac{15+x + 15-x}{(15-x)(15+x)} \right) = 30 \left(\frac{30}{225-x^2} \right)$$

$$\frac{9 \times 81}{60 \times 36 \times 20 \times 10} = \frac{1}{225-x^2} = \frac{1}{200} = \frac{1}{225-x^2}$$

$$\Rightarrow 225 - x^2 = 200$$

$$x^2 = 25$$

$$x = 5$$

10) A boat travels 10 km upstream and 5 km downstream in 3 hrs. If the speed of boat in still water is 7 kmph. Find the speed of stream.

Ans: 3 kmph

$$d = 15 \text{ km}$$

$$U = 7 - x$$

$$d = 10 \text{ km upstream} + 5 \text{ km downstream} = 15 \text{ km}$$

$$D = 7 + x$$

$$t = t_u + t_d$$

$$3 = \frac{10}{7-x} + \frac{5}{7+x} \Rightarrow 3 = \frac{70 + 10x + 35 - 5x}{49 - x^2}$$

$$147 - 3x^2 = 105 + 5x$$

$$3x^2 + 5x - 42 = 0$$

$$3x^2 + 14x - 9x - 42 = 0$$

$$x(3x + 14) - 3(3x + 14) = 0$$

$$x = 3$$

11. The speed of boat in still water is 15 kmph. And speed of stream is 3 kmph. Find the distance travelled downstream in 12 mins.

$$\text{Ans: } 3.6 \text{ km}$$

$$S_b = 15 \text{ kmph} \quad S_s = 3 \text{ kmph}$$

$$D = S_b + S_s = 18 \text{ kmph}$$

$$60 \text{ min}$$

$$1 \text{ hr} \rightarrow 18 \text{ km}$$

$$12 \text{ min} \rightarrow x$$

$$\Rightarrow \frac{18}{60} \times 12 = 3.6$$

12. A boat takes half time in moving to a certain distance downstream than upstream. Find the ratio of speed of boat in still water and to that of speed of current.

$$\text{Ans: } 3:1$$

$$U = 2 \text{ kmph}$$

$$D = 2x \text{ kmph}$$

$$t_D = \frac{1}{2} t_u$$

$$8x \frac{1}{t}$$

$$S_b = \frac{D+U}{2} = \frac{3x}{2}$$

$$S_s = \frac{D-U}{2} = \frac{x}{2}$$

$$S_b : S_s = \frac{3x}{2} : \frac{x}{2}$$

$$= 3:1$$

- 13) A rows to a place 48 km distance and comes back in 14 hrs. He finds that he can row 4 km with the stream in the same time as 3 km against the stream. Find the speed of the stream.

$$\text{Ans: } 1 \text{ kmph}$$

$$S \propto d$$

$$\frac{U}{D} = \frac{3}{4}$$

$$U = 3x = 6 \text{ kmph}$$

$$D = 4x$$

$$t = t_u + t_D$$

$$14 = \frac{48}{3x} + \frac{48}{4x}$$

$$\frac{2}{x} = 1$$

$$x = 2$$

$$S_b = \frac{D+U}{2} \quad [\because D = 4(2) = 8 \text{ kmph}]$$

$$U = 3(2) = 6 \text{ kmph}$$

$$= \frac{8-6}{2}$$

$$= \frac{2}{2} = 1$$